

Vutukuri Sathvik

FULL-STACK ENGINEER · BACKEND & DISTRIBUTED SYSTEMS

11+ years as a **full-stack engineer** building complete products end to end — from large-scale **backend and distributed systems** in Java (high-throughput messaging, distributed scheduling, decisioning infrastructure) to the **React** web portals and dashboards on top. Deepest on the systems side, owning architecture through **production delivery** at scale (platforms serving 1B+ requests/day), and comfortable across the full stack.

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EXPERIENCE

Member of Technical Staff I · Backend Architect PayPal

Bengaluru, IN
Apr 2022 – Present

🧠 KAIROS — Communication Decision Platform

- ▶ Designed core components of **KAIROS**, PayPal's Communication Decision Platform — the centralized layer that governs what every consumer is communicated, when, and on which channel — designed, delivered, and running in production.
- ▶ Designed the **Intent Management System (IMS)** — the ingestion and management layer for decision intents flowing into the platform from producing systems.
- ▶ Architected the **Deferred-Intent Scheduler** — a durable, leaderless distributed timer on **Google Bigtable + Pulsar** sized for ~25–30M deferred intents/day with bursts up to 2M due at once. Used per-row atomic **compare-and-set (checkAndMutate)** claims for single-fire dispatch and **Kafka consumer-group** hash-partitioning for leaderless work distribution across autoscaling instances.
- ▶ Owned the Scheduler's end-to-end low-level design: hash-prefixed row-key schema for even node distribution, scan-claim-dispatch pipeline, cell-timestamp-based garbage collection, failure recovery, and at-least-once dispatch with idempotent re-evaluation.
- ▶ Evaluated alternative Bigtable data models (row-per-intent vs wide-row bucketing) with DBA architects, selecting the design that preserved write parallelism at burst scale.
- ▶ Designed part of the **Ledger** component and defined the ingestion/dispatch contracts integrating the Scheduler with the Decision Engine's trigger pipeline.

⚡ PayPal Communications — Comms Platform

- ▶ Architected and own the end-to-end backend platform powering all PayPal communications — sending, scheduling, and delivering across **Push, Notification Center, Critical Alert & Silent Push** for PayPal, Venmo & Xoom tenants.
- ▶ Driving the **communications platform revamp** — redesigning service architecture for scalability, multi-region support, and integration with the centralized decisioning layer.
- ▶ Designed and built **Beacon** — the production web portal for PayPal Communications used company-wide to schedule and manage communications; owned it end-to-end (React frontend).
- ▶ Made the comms platform a **producer of intent objects** so every send/suppress/defer decision is centrally evaluated against eligibility, ML scoring, cap enforcement & quiet hours.

📄 Verse — Long-Form Content Platform

- ▶ Designed the backend and end-to-end flow for **Verse**, a modular, partly server-driven platform that lets product teams ship content experiences (FAQs, T&Cs, education pages) without building custom UI each time.
- ▶ Architected it for **multi-tenant** use across PayPal, Venmo & Xoom via a shared core with tenant-specific modules, driving content and predefined design-system templates from the server with context-aware rendering and offline-capable caching.
- ▶ Built the backend for the admin tool and defined the client SDK contract — single-call integration, deep-linkable pages, and multiple presentation modes.

🏠 Server-Driven UI (SDUI) Platform

- ▶ Led backend for PayPal's in-house **Server-Driven UI platform** — built from scratch and shipped to production, serving **1 billion+ requests/day** and powering core PayPal app & web experiences.
- ▶ Architected core SDUI services: version management, page inheritance, schema versioning & database design — enabling consistent UI delivery across iOS, Android & Web.

🏆 PayPal SPOT Awards — Q3 2022 & Q4 2023

SKILLS

LANGUAGES

Java (8–25) JavaScript
SQL

DISTRIBUTED SYSTEMS

System Design
Distributed Scheduling
Event-Driven
Consistency & Idempotency
High Throughput
Fault Tolerance

CONCURRENCY

Virtual Threads
Structured Concurrency
CompletableFuture
Multithreading

BACKEND

Spring Boot GraphQL
REST Microservices
gRPC

MESSAGING & STREAMING

Kafka Pulsar

DATA & STORAGE

Bigtable Oracle
MongoDB

CLOUD & INFRA

GCP GKE Kubernetes
Docker Amazon S3

FRONTEND / FULL-STACK

React JS Redux
JavaScript Tailwind
HTML/CSS
Admin Dashboards

OBSERVABILITY

Datadog Grafana
Prometheus Micrometer
OpenTelemetry

- Senior Software Engineer

Informatica

Architected and built **SecureStats** end-to-end — security-flaw management platform with full backend, frontend, database design & APIs.

Co-designed **Flaw Management Service & Notification Service** in a microservices architecture.

Mentored 5 engineers on service design & architecture decisions. **INFASTAR Award Q1 2021.**

Bengaluru, IN

Mar 2020 – Apr 2022

- Senior Software Engineer

Investnet Yodlee

Architected **Juggernaut** — microservice platform for automated issue resolution on Yodlee PFM. Owned end-to-end backend & frontend design.

Designed observability stack with Spring Boot Actuator, Micrometer, Prometheus & Grafana; microservice infra with Eureka, Zuul & Hystrix.

Yodlee Dream Team Award — Juggernaut R&D (Q2 2018, Q2 2019).

Bengaluru, IN

Nov 2015 – Mar 2020

SELECTED TECHNICAL HIGHLIGHTS

- ▶ **Distributed scheduling at scale** — designed a durable, leaderless timer on Bigtable + Pulsar for ~25–30M deferred intents/day with 2M-at-once bursts, using per-row atomic compare-and-set for single-fire dispatch without a coordinator.

▶ **High-throughput serving** — led the backend for a Server-Driven UI platform serving **1B+ requests/day** across iOS, Android & Web.

▶ **Schema & storage design** — evaluated competing Bigtable data models with DBA architects and chose the design that preserved write parallelism under burst load.

▶ **Messaging & event-driven systems** — built platforms on Kafka and Pulsar spanning ingestion, scheduling, and dispatch, with idempotent, at-least-once delivery semantics.

▶ **Full-stack delivery** — built the React web portals and admin dashboards on top of the backend systems (e.g. Beacon), owning features from data model to UI.

▶ **End-to-end ownership** — consistently took systems from problem statement and architecture through low-level design, implementation, and production rollout.

FOUNDATIONS

DSA OOP

Design Patterns

EDUCATION

B.Tech — CSE
JNTU Kakinada · GPA 7.9/10
2011 – 2015

AWARDS

🏆 **PayPal SPOT Awards**
SDUI · Q3 2022, Q4 2023

★ **Informatica INFASTAR**
SecureStats · Q1 2021

🏆 **Yodlee Dream Team**
Juggernaut R&D · Q2 2018, Q2 2019

🌟 **Yodlee Spot Awards**
Q2/Q3 2016 · Q1 2018 · Q3 2019